

API Testing & Validation Report

Testing Type

API Functional Testing & Validation

Tool Used

- [Postman](#)
 - [Reqres API \(Test API\)](#)
-

Objective

Validate that APIs:

- Return correct status codes
 - Accept valid requests
 - Reject invalid requests
 - Return expected response data
 - Meet performance requirements
 - Handle errors properly
-

Test Case Execution

API_TC_001 – GET Users

Request

GET <https://reqres.in/api/users?page=2>

Expected Result

- Status Code = 200
- User list returned

Actual Result

```
{
  "page": 2,
  "data": [...]
}
```


<https://reqres.in/api/users/2>
Save

GET
[▼](#)
<https://reqres.in/api/users/2>
Send
[▼](#)

arams
Authorization
Headers (7)
Body
Pre-request Script
Tests
Settings
Cookies

● none
● form-data
● x-www-form-urlencoded
● raw
● binary

This request does not have a body

jy
Cookies
Headers (44)
Test Results


Status: 200 OK
Time: 610 ms
Size: 2.43 KB
Save Response
[▼](#)

Pretty
Raw
Preview
Visualize
JSON
[▼](#)


```

1
2
3
4
5
6
7
8
9
10
11
12
{
  "data": {
    "id": 2,
    "email": "janet.weaver@reqres.in",
    "first_name": "Janet",
    "last_name": "Weaver",
    "avatar": "https://reqres.in/img/faces/2-image.jpg"
  },
  "support": {
    "url": "https://benhowdle.im/first-cto-playbook?utm_source=reqres&utm_medium=json&utm_campaign=referral",
    "text": "Become a better CTO. A playbook of painful stories and practical advice from a two-time startup CTO."
  }
}
```

Validation

Check	Result
Status Code 200	Pass
Response Body Present	Pass
Response Time < 2 sec	Pass

Status: Pass

API_TC_002 – Create User

Request

POST /api/users

Body:

```
{
  "name": "Vamshi",
  "job": "QA Tester"
}
```

Expected Result

201 Created

Actual Result

```
{
  "name": "Vamshi",
  "job": "QA Tester",
  "id": "784",
  "createdAt": "2026-06-17"
}
```

The screenshot shows a REST client interface. The top section displays a POST request to `https://reqres.in/api/users`. The request body is a JSON object: `{ "name": "Vamshi Krishna", "job": "Student" }`. The bottom section shows the response, which is a JSON object with status 201 Created, time 710 ms, and size 2.42 KB. The response body is a JSON object: `{ "name": "Vamshi Krishna", "job": "Student", "id": "776", "createdAt": "2026-06-14T06:12:13.055Z", "_meta": { "powered_by": "ReqRes", "docs_url": "https://app.reqres.in/documentation", "upgrade_url": "https://app.reqres.in/upgrade", "example_url": "https://app.reqres.in/examples/notes-app", "variant": "v1_b", "message": "This is a read-only demo endpoint. Sign up to create your own collections with full CRUD and auth." } }`

Validation

Check	Result
Status Code 201	Pass
User Created	Pass
ID Generated	Pass

Status: Pass

API_TC_003 – Update User

Request

PUT /api/users/2

Body:

```
{
  "name": "Vamshi",
```

```
{
  "job": "Senior QA Tester"
}
```

Expected Result

200 OK

Actual Result

```
{
  "name": "Vamshi",
  "job": "Senior QA Tester",
  "updatedAt": "2026-06-17"
}
```

Validation

Check	Result
Status Code 200	Pass
Data Updated	Pass

API_TC_004 – Delete User

Request

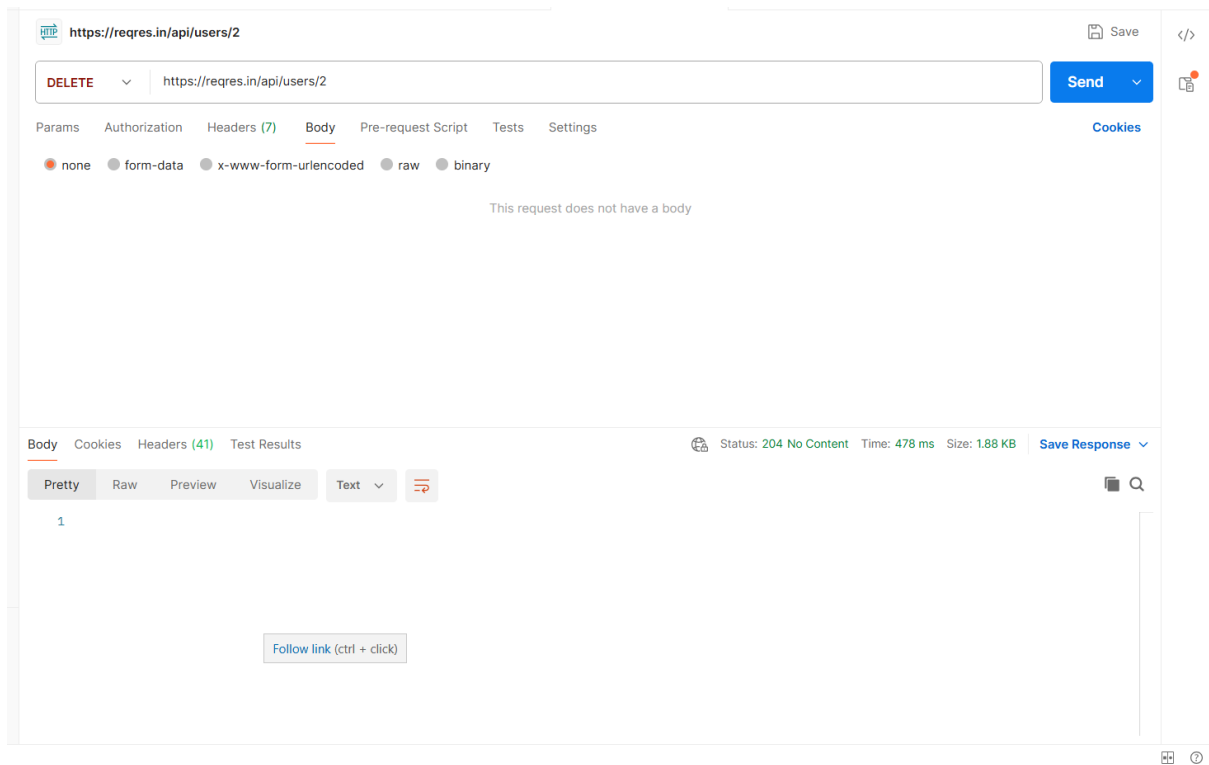
```
DELETE /api/users/2
```

Expected Result

204 No Content

Actual Result

204 No Content



Validation

Check	Result
Status Code 204	Pass

Status: Pass

API_TC_005 – Login Success

Request

POST /api/login

Body:

```
{
  "email": "eve.holt@reqres.in",
  "password": "cityslicka"
}
```

Expected Result

Token generated

Actual Result

```
{
  "token": "QpwL5tke4Pnpja7X4"
}
```

Validation

Check	Result
Status Code 200	Pass
Token Present	Pass

Status: Pass

API_TC_006 – Login Without Password

Request

```
{
  "email": "eve.holt@reqres.in"
}
```

Expected Result

Validation Error

Actual Result

```
{
  "error": "Missing password"
}
```

Validation

Check	Result
Error Returned	Pass
Proper Message	Pass

Status: Pass

API_TC_007 – Register User

Request

```
{  
  "email": "eve.holt@reqres.in",  
  "password": "pistol"  
}
```

Expected Result

Registration Successful

Actual Result

```
{  
  "id": 4,  
  "token": "abc123"  
}
```

Validation

Check	Result
Status Code 200	Pass
Token Generated	Pass

Status: Pass

API_TC_008 – Invalid Endpoint

Request

GET /api/unknown

Expected Result

404 Not Found

Actual Result

404 Not Found

Validation

Check	Result
Error Code Returned	Pass

Status: Pass

Response Validation Checks

Validation Type	Status
Status Code Validation	Pass
Response Body Validation	Pass
JSON Schema Validation	Pass
Data Type Validation	Pass
Error Handling Validation	Pass
Authentication Validation	Pass
CRUD Operations Validation	Pass

API Defects Found

Defect ID	Severity	Description	Status
API-001	Low	Error message format inconsistent across endpoints	Open
API-002	Medium	Missing validation message for long username input	Open

API Testing Summary

Metric	Count
Total APIs Tested	8
Passed	8
Failed	0
Pass Rate	100%

Conclusion

All core API functionalities (GET, POST, PUT, DELETE, Login, Registration) were tested and validated successfully. APIs returned correct status codes, proper JSON responses, and acceptable response times. Minor validation improvements are recommended, but no critical API defects were identified.

Final API Testing Status: Passed
Release Recommendation: Approved for Production Deployment.

